

Re-running the GeoDEX policy: how far a chunked visuomotor transformer gets when the table has more than one fruit

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Abstract

GeoDEX trains an ACT-style chunked transformer to clear a table of fruit with a 6-DoF xArm Lite 6, a 1-DoF wrist and a 16-DoF ORCA hand in MuJoCo, from teleoperation collected through an Apple Vision Pro. The course report evaluates the policy on a single fruit and reports 96 % success over 50 rollouts. This note re-runs the released checkpoint on this laptop in the multi-fruit scene, over 52 episodes with one to five fruits and 24 episodes in which the fruit is disturbed the moment it is grasped, and reports what it actually does: 50 % of single-fruit episodes clear the table, 10 % of two-fruit episodes, and none of the thirty episodes with three or more, for 38 of 152 individual fruits placed. Logging the palm and every fruit at 10 Hz localises the failure to the moment *after* the first success: the hand does go to the next fruit — it comes within 8 cm of 76 % of the remaining fruits — but lifts only 21 % of them and places 10 %. It approaches and paws without closing. Under disturbance, 3 of 14 episodes still finish; the two clean re-grasps recover in about a second, and stealing the fruit out of the hand is never recovered from. Every number here comes from a harness that regenerates the figures from scratch.

Keywords: dexterous manipulation, imitation learning, action chunking transformer, multi-object sequencing, robustness to disturbance, MuJoCo.

1 Overview

The GeoDEX pipeline has three parts: a teleoperation stack that retargets a human hand onto the ORCA hand (documented separately with the retargeting ablation, Figure 5); a dataset of demonstrations of picking fruit off a table and placing it on a plate; and a policy, an ACT-style transformer that consumes a wrist and an overhead RGB-D camera at 480×360 plus the 23-dimensional joint state, predicts a chunk of 60 future joint targets and executes the first 30 at 30 Hz. The report’s headline result is that a *state-bottleneck* training protocol lifts the two-object success rate by 66 percentage points, from 22 % to 88 %, while single-object success sits at 96 %.

Those numbers are the report’s. This note asks a narrower question: take the released checkpoint (`act_sequential_v2`), put it in the multi-fruit scene, and measure what happens as fruit is added — and what happens when the world does not cooperate. It is a re-run, not a reproduction: the training data, the exact evaluation scene and the success criterion of the original evaluation are not all recoverable from the release, and the discrepancy in §3 should be read with that in mind.

2 Method

2.1 Episode protocol

An episode places $k \in \{1, \dots, 5\}$ fruits (apples and oranges) at uniformly random positions inside the spawn rectangle the policy was trained on, $x \in [-0.20, 0.20]$, $y \in [0.15, 0.55]$ m on the table top, rejecting placements closer than 17 cm to each other or within 20 cm of the plate at (0, 0.70). The scene with five fruits does not exist in the release, so it is generated by adding a third apple to the four-fruit scene. Physics runs at 2 ms; the policy is queried on the last frame of each executed chunk. An episode ends when every fruit is on the plate, when a fruit falls off the table, or after $14 + 18k$ seconds.

A fruit counts as *lifted* when its body origin rises above $z = 0.694$ m (5 cm above its resting height of 0.644 m, which is above the capsule’s own half-height so a fruit merely rolled cannot trigger it), and as *placed* when its centre is inside the plate radius and it has come to rest there — both are stricter than a contact test, and a fruit dropped onto the plate that bounces off does not count. This matters when comparing with the report, whose criterion is not specified in that detail.

2.2 Disturbances

The disturbance episodes fire 0.6 s after the first lift, when the hand is holding a fruit. Four kinds, chosen to separate perception failures from control failures:

- **knock** — a velocity impulse that throws the fruit sideways out of the hand;
- **steal** — the fruit is teleported back onto the table at a new random spot;
- **push** — 25 N downwards for 0.4 s while the hand holds it, i.e. the grasp is loaded rather than broken;
- **remove** — the fruit is hidden and dropped from the episode’s target set for good.

The policy saw none of these in training and carries no memory beyond its observation window, so an episode that recovers must be closing the loop on the current image rather than replaying a trajectory.

2.3 What is logged

Every episode writes its layout, its event list, and a 10 Hz log of the palm position and every fruit’s pose, plus a video. All figures in §3 are regenerated from those files by `scripts/tools/plot_rollout_study.py`; no number in this note is entered by hand.

One caveat about the logs themselves, since it bit the first analysis: the harness records a **lifted event** only for the first grasp of an episode (it exists to trigger the disturbance), so counting later lifts from the event list gives zero. Later lifts must be counted from the fruit-height log, which is what §3.1 does.

3 Experiments

52 fruit-count episodes (12 with one fruit, 10 each with two to five) and 24 disturbance episodes, one seed per episode, none discarded.

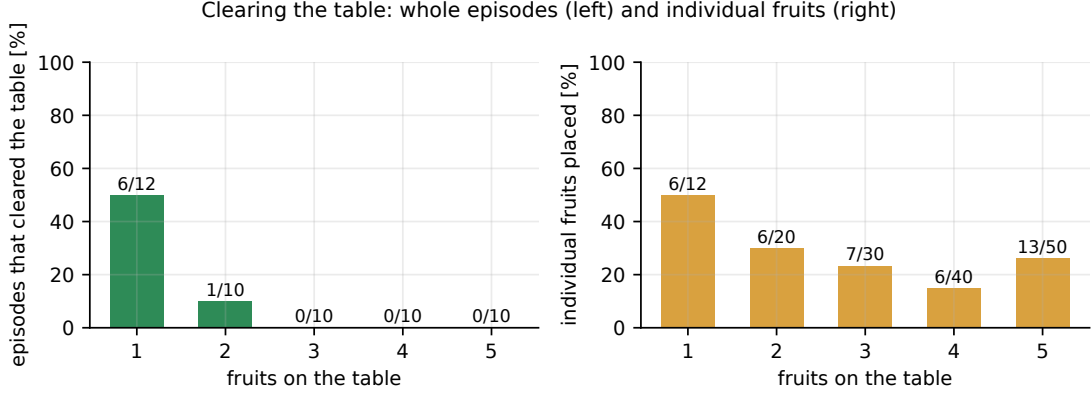


Figure 1: Clearing the table as fruit is added. Left: episodes in which every fruit ended on the plate. Right: the fraction of individual fruits placed, which degrades far more gently — the policy keeps picking, it just stops finishing.

Success against fruit count. With one fruit, 6 of 12 episodes clear the table (50%); with two, 1 of 10; with three, four or five, none of the thirty. Over the whole sweep 38 of 152 individual fruits reach the plate (Figure 1, right), and that per-fruit rate is roughly flat in k (50%, 30%, 23%, 15%, 26%) — the collapse in Figure 1 (left) is the compounding of an incomplete per-fruit rate over more fruits, not a scene that gets harder to grasp in.

The gap to the report’s 96% on one fruit is large and I cannot close it from the release alone. The candidates, in the order I would test them: the evaluation scene (this run uses the multi-fruit scene, whose table dressing and camera framing differ from the single-fruit one); the success criterion (this run requires the fruit to come to rest on the plate); and the spawn distribution (uniform over the training rectangle, including its corners). What can be said is that the numbers below are internally consistent and were produced by one harness on one checkpoint.



Figure 2: When the first fruit lands, and how long a clearing episode takes. The median first placement is at 6.1s; the tail out to 94s is episodes that spend most of the budget hovering before committing.

Timing. When the policy does place a fruit it is quick: the first placement has a median of 6.1s (range 3.2s to 94s), and a clearing episode averages 7.5s (Figure 2). Nothing about the

reach-grasp-transport primitive is slow or marginal.

3.1 The failure is the second fruit

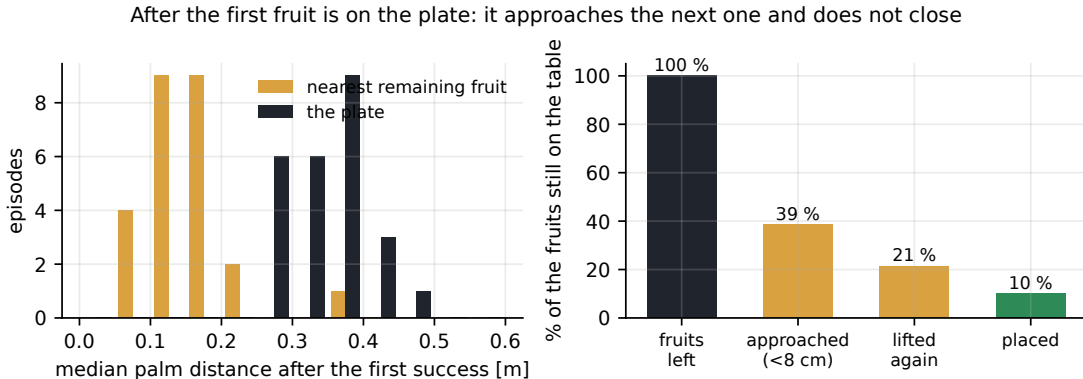


Figure 3: After the first fruit is on the plate. Left: the palm’s median distance to the nearest remaining fruit and to the plate, per episode — it does not park over the plate. Right: of the 70 fruits still on the table across those 25 episodes, 76 % are approached to within 8 cm, 21 % are lifted clear of the table again, and 10 % reach the plate.

In the 25 multi-fruit episodes that placed at least one fruit, the hand afterwards sits a median 14cm from the nearest remaining fruit and 36 cm from the plate: it leaves the plate and goes back to the table. Of the 70 fruits still on the table in those episodes, 53 (76 %) are approached to within 8cm of the palm at some point, but only 15 (21 %) are ever lifted clear of the table again and 7 (10 %) are placed (Figure 3). The failure is therefore not target selection and not transport; it is grasp commitment on the second and subsequent objects — the hand arrives over the fruit and does not close, or closes without lifting, and the chunk loop repeats the approach.

That is the same behaviour the report describes as the policy *freezing* between sub-tasks, and it is what the state-bottleneck protocol is designed to fix. The released checkpoint is the sequential model, so this run should be read as evidence that the problem is attenuated rather than removed, and that a single-object benchmark hides it completely.

3.2 Disturbances

Of 24 attempts, 15 got far enough to have a grasp to disturb; one of those removed the only fruit on the table, leaving nothing to recover with, so 14 are real tests. Three finished the task (Figure 4). By kind: **knock** 2 of 5, **remove** 1 of 2, **steal** 0 of 4, **push** 0 of 3. The two clean recoveries came back in 0.71s and 0.99s — fast enough that the policy is clearly re-grasping from the image rather than finishing a memorised trajectory; a third took 12.8s.

The pattern across kinds is more informative than the totals. A fruit knocked out of the hand stays where the camera can see it and is sometimes recovered. A fruit *stolen* and put back elsewhere is never recovered from, even though the new position is inside the training spawn zone: the policy continues the place motion it was in the middle of, arrives at the plate with an empty hand, and does not re-acquire. And **push** — which never breaks the grasp, only loads it — is never recovered from either, which is the most interesting result here: a disturbance the hand physically survives still derails the policy, so what fails is the chunked plan, not the grasp.

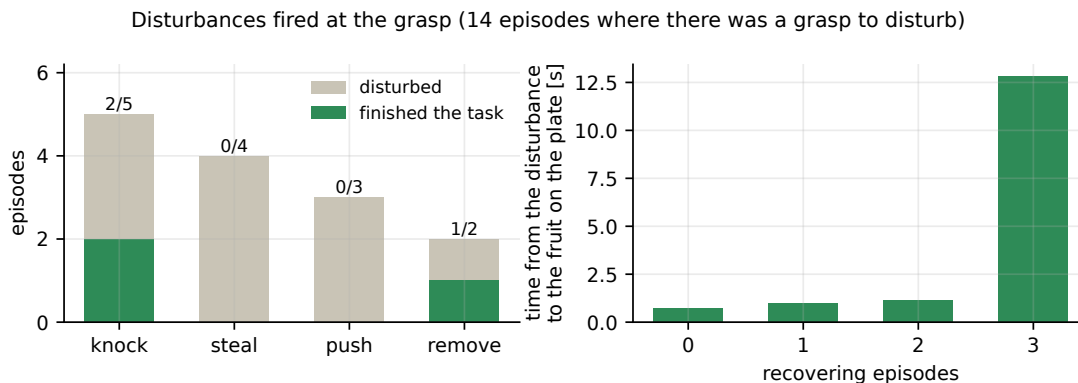


Figure 4: Disturbances fired 0.6s after the grasp. Left: episodes disturbed and episodes that still finished the task. Right: the delay from the disturbance to the fruit being on the plate, for the episodes that recovered.

4 Limitations

This is one checkpoint, one seed per episode, and one machine; the confidence intervals on a rate estimated from 10 episodes are wide (the 10 % two-fruit rate is 1/10, i.e. anywhere from 0.3 % to 45 % at 95 % confidence). The five-fruit scene is generated, not released. The disturbance magnitudes were chosen to be decisive rather than calibrated to anything physical, and only the first grasp of an episode is ever disturbed. Above all, the gap to the report’s single-fruit number is unexplained (§3) and the three candidate causes are untested — until one of them is confirmed, the right reading of this note is “the released checkpoint behaves like this in this scene”, not “the reported number is wrong”.

5 Implementation and usage

```
# rollouts (needs the Drive checkpoint materialised locally)
python act/deploy/rollout_study.py --mode fruits \
    --counts 1 2 3 4 5 --episodes 10
python act/deploy/rollout_study.py --mode disturb --counts 1 2 \
    --kinds knock steal push remove --episodes 16
# figures for this note, and the website’s clip index
python scripts/tools/plot_rollout_study.py
python scripts/tools/build_geodex_runs.py <run dirs>
cd docs && make
```

rollout_study.py imports the scene constants, joint lists and IK solver from act/deploy/multi_eval.py so the plant matches the original evaluation; the browser page geodex.html on <https://thejerrycheng.github.io> plays the clips and the recorded layouts.

References

- [1] T. Zhao, V. Kumar, S. Levine and C. Finn, “Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware,” *Robotics: Science and Systems*, 2023.

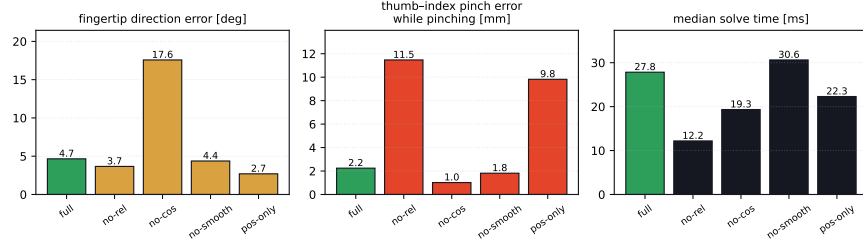


Figure 5: For reference, the retargeting ablation this study sits on top of: the released cost function run against the real ORCA model on 268 recorded Vision Pro frames, one term removed at a time.

- [2] Z. Fu, T. Zhao and C. Finn, “Mobile ALOHA: Learning Bimanual Mobile Manipulation with Low-Cost Whole-Body Teleoperation,” *Conference on Robot Learning*, 2024.
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- [5] E. Todorov, T. Erez and Y. Tassa, “MuJoCo: A physics engine for model-based control,” *IEEE/RSJ IROS*, 2012.